

Science · LAUNCH · Autumn 1 · Week 08 · SCI-LAUNCH-A1-W08-L1 · Date _____ Enzyme Action: Preparing the pH Investigation Today I can... ☐ Use enzyme, substrate, active site and specificity accurately in an explanation ☐ Explain how an unsuitable pH can change an active site's shape and reduce enzyme activity ☐ Identify pH as the independent variable, time to starch disappearance as the dependent variable, and suitable control variables Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R _____	
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Enzyme Action: Preparing the pH Investigation · Route card

Reference only · use the task and sentence stem in your book.

Arrival · four short answers in your book

1. Which structure holds chromosomes in a eukaryotic animal cell?
2. Link to today: In diffusion, do particles move from higher to lower concentration or the reverse?
3. Which substance moves through a partially permeable membrane in osmosis?
4. Which transport process uses energy to move particles against a concentration gradient?

Starter

Which iodine observation shows starch is still present: blue-black or orange-brown?

Exit ticket

Under the heading Exit ticket: answer the questions in your book.

1. Explain why amylase acts on starch and can be used again.
2. Name the IV, define the timed DV with units, and name one control with a reason.

Independent

1. Write a testable prediction for how pH could affect amylase acting on starch. Explain why the enzyme can act again. [2 marks]
2. Name the IV and define the timed DV with units. Give three controls and justify one. [6 marks]
3. Order the six method cards and explain why iodine is used to test separate sampled drops. [2 marks]
4. Distinguish a no-active-amylase control sample from a control variable. Predict the control result. [3 marks]
5. State suitable table headings for pH, trial, endpoint time / s, observation and relative rate $1/\text{time} / \text{s}^{-1}$. State what to record if there is no endpoint by 300 s. [3 marks]

Supported

1. Use the model to finish: "Amylase is the _____. Starch is the _____. The enzyme is not _____."
2. Match pH / time to endpoint / temperature to CHANGE / MEASURE / KEEP THE SAME. Give one reason for keeping temperature the same.
3. Order method cards A–F on the reference page. Point to the actions done before mixing.
4. Point to the endpoint sample. Predict the iodine result in a separate sample with starch but no active amylase.
5. Use these results headings: pH | trial | endpoint time / s | observation. If no endpoint is seen by 300 s, choose "0 s" or "no endpoint by 300 s".

Choice

1. Choose the catalyst: amylase / starch / iodine. Predict: pH 4 will reach the endpoint sooner / later than pH 7. Give one active-site reason. Complete: The enzyme can act again because it is not _____. Record C beside your book response.
2. Match pH / endpoint time / temperature to change / measure / keep the same. Give one reason. Order the six supplied method cards. Choose the iodine check: separate sampled drops / the reaction mixture. Choose no active amylase as a comparison group / changed pH; predict blue-black / orange-brown.

3. Choose the change heading: pH / temperature; repeat heading: trial / colour; observation heading: colour / pH. Choose time unit: s / s⁻¹; rate unit: s⁻¹ / s. Choose the truthful entry at 300 s: zero / no endpoint by 300 s. Give one reason.

Aspire

1. Predict a testable pH–activity pattern. Explain specificity and reuse without claiming that reduced activity proves permanent denaturation. [4 marks]
 2. Specify the IV, operational DV and four controls. Justify two controls against competing explanations. [8 marks]
 3. Order method cards A–F; explain why iodine goes into separate check drops and why a consistent start matters. [3 marks]
 4. Distinguish the no-active-amylase control from a control variable. What would an unexpected orange-brown result make you check? [3 marks]
 5. State the evidence from illustrative samples: blue-black at 90 s, orange-brown at 120 s, sampled every 30 s. State the recorded time, the interval containing the threshold crossing, and one precision improvement. For another run with no endpoint by 300 s, choose a truthful table entry. [4 marks]
- Staff mark I, S or C beside this task.

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LAUNCH · GCSE Biology · Enzyme Action: Preparing the pH Investigation

LAUNCH SCIENCE | last lesson |

Enzyme action and pH

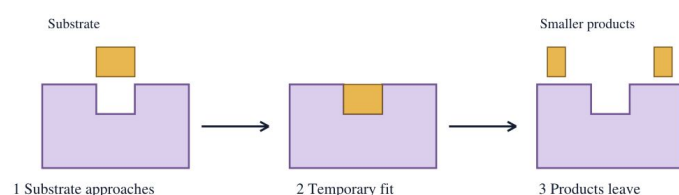
Knowledge organiser

Learning objective

I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely.

Success criteria

1. Use enzyme, substrate, active site and specificity accurately in an explanation.
2. Explain how an unsuitable pH can change an active site's shape and reduce enzyme activity.
3. Identify pH as the independent variable, time to starch disappearance as the dependent variable, and suitable control variables.



Active site: a specific region where substrate binds. Enzyme remains available.

A complementary active site

Specificity means acting on particular substrates. Amylase catalyses starch breakdown into smaller sugars. Products leave; the enzyme remains available. This simplified shape model does not show literal rigid molecules.

Connect pH to a measured endpoint

pH describes acidity or alkalinity. Unsuitable pH can alter the active site and reduce activity. Denaturation disrupts the enzyme's three-dimensional structure and function. Slower activity alone does not prove permanent denaturation.

Plan a valid comparison

Change pH. Time the first sampled drop in which iodine remains orange-brown (starch not detected). Blue-black means starch detected. Match temperature, concentrations, volumes and sampling. check separate drops; keep iodine out of the main mixture.

Read the evidence honestly

A no-active-amylase comparison should remain starch-positive. Record "no endpoint by 300 s" when that is all that was observed; do not invent zero time or zero rate. The optimum pH is to be investigated.

Made by Matt | Use the reference to explain, calculate and evaluate.

Enzyme Action: Preparing the pH Investigation

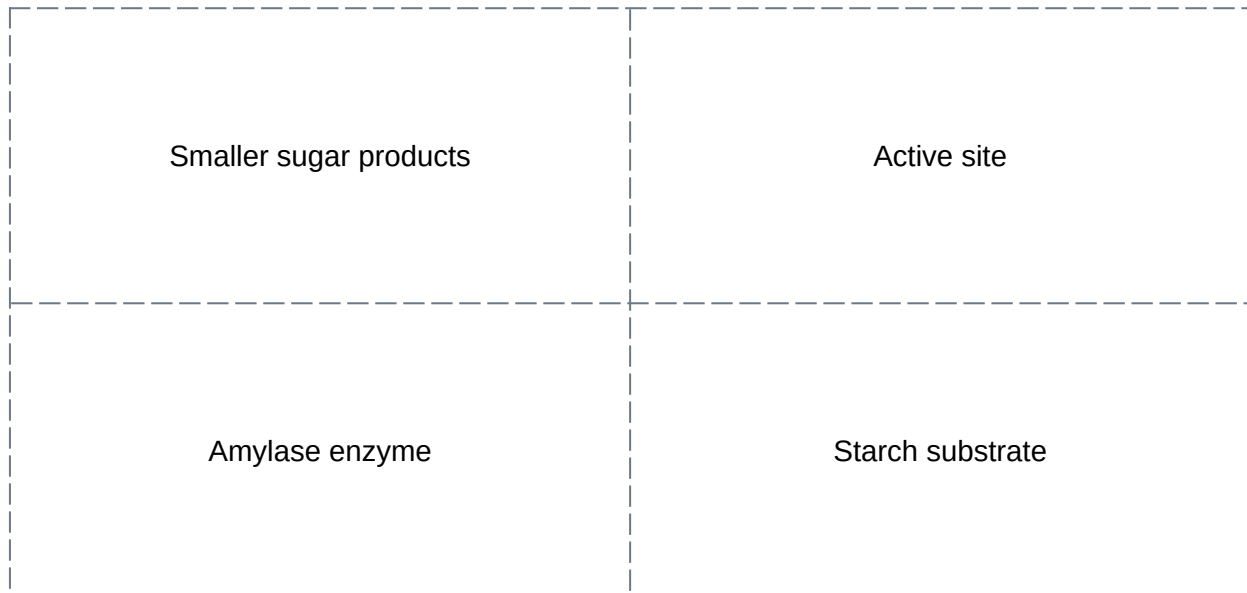
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Staff: cut these apart before the lesson.

Move these pieces for the task. Record the answer and reason in your book.

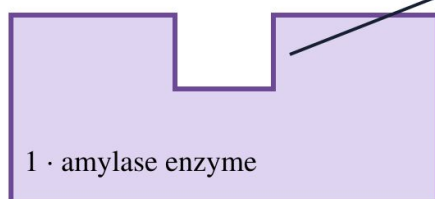


Enzyme model

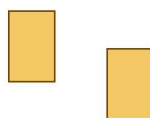
2 · starch substrate



3 · active site



1 · amylase enzyme



4 · smaller sugar products